

An analysis of the persistent availability of unmarked retracted articles in academic social networks

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Abstract— Objective: This study aimed to evaluate the prevalence of unmarked retracted publications across two major research-sharing platforms. **Design:** In this cross-sectional study, the Retraction Watch database on retractions was searched for papers fulfilling the following criteria: From Brazil, in all areas of health science, published between 2010 and 2020, and retracted for unreliability-related reasons (data fabrication, fraud, and data/methods/analysis problems). Expressions of Concern (EoCs) were also included. Once selected, these articles were searched on two research-sharing platforms: Research Gate and Academia.edu. Searches were conducted between Nov/24 and Jan/25, and papers were considered available when the manuscript could be found with no indication that a retraction/EoC had occurred and no retracted/EoC version could be found in the respective platform. Identified “abstracts only” within the conditions were also recorded, and, in addition, a standard Google Search was conducted, looking for papers/abstracts available in other platforms. All searches were performed by two researchers and independently double-checked by a third one. In cases of doubt, researchers convened to meet a consensus. The following information was collected: date of publication, date of retraction, title and (Google Scholar) citations received. **Results:** A total of 66 retractions could be selected. Out of these, 22 (1/3) were available unmarked at the platforms (13 Academia, 5 Research Gate, 4 in both). Additionally, 07 other unmarked abstracts were found (04 Research Gate, 03 in other repositories). The total post-year of retraction citations for the full papers was 730 (average 33.2), and for the abstracts, 166 (average 23.7). **Conclusions:** This study provides empirical evidence for the high persistence of unmarked retracted publications in academic social networks, thus highlighting potential noteworthy gaps in retraction-signaling practices on academic social networks.

Keywords— Retraction, scientific publications, citation, research ethics

I. INTRODUCTION

The retraction of scientific articles is a process in which a previously published article is invalidated due to serious problems that compromise its integrity, namely fraud, plagiarism, or significant methodological errors [1, 2]. These

retractions serve as a crucial mechanism for maintaining research integrity within the scholarly ecosystem [3]. However, the efficacy and quality of retraction notifications across academic platforms remain unsatisfactory [3,4], potentially compromising the self-correcting nature of science [5].

The use of such papers can have consequences such as loss of scientific credibility, decline of science respect for ethical precepts, and the negative impacts of projects based on erroneous information. Thus, the fact that articles remain in databases without a clear indication of their invalidation is a cause for concern in the scientific community [6-8], which is investing in measures to deal with this problem.

Despite the rising interest in the problem, there are still significant gaps in understanding its causes. So, the extent of the presence of articles without proper retraction signposting in Internet databases is not fully understood, and neither are the reasons for their continued citation [3,4].

Therefore, the aim of this study was to identify retracted articles available on internet databases and to verify their citation rates, with the aim of contributing to an understanding of the impact of this problem and the development of more effective policies for controlling and signaling retractions.

II. METHODS

A. Data

In this descriptive study, the Retraction Watch (RW) database on retractions was searched for papers fulfilling the following criteria: from Brazil, in all areas of health science, published between 2010 and 2020, and retracted for unreliability-related reasons. Nationality was defined according to the first author’s affiliation. Unreliability was defined by alleged retraction reasons related to data fabrication, fraud, and problems with data/methods/analysis.

Expressions of Concern (EoCs) were also included in the search. These are warnings, usually implemented when a paper is under investigation for retraction but no decision

has yet been found. They indicate serious concerns on the validity of a paper and thus should also impede paper citation. When a paper received a retraction after an EoC only the retraction was considered for analysis.

B. Search criteria

Once selected, the RW-identified retractions were searched on two research-sharing platforms. The first, Research Gate, has about 20 million members from 190 countries and presents itself as “the Facebook of scientists”. The second, Academia.edu, claims to have more than 280 million users [9.10]. These are the two most prominent research-sharing platforms used by researchers.

Searches were conducted in Nov/24 and repeated in Jan/25, and papers were considered available when the manuscript could be found with no indication that a retraction/EoC had occurred and no retracted/EoC version could be found in the respective platform. Searching was performed with the standard search engines of the platforms, by looking for the article title, and, also, for the title in conjunction with the keywords “retraction”, “retracted”, “withdrawn” and “expression of concern” (in order to make sure that a retracted version was not available). In addition, once identified, a double-check was performed, so as to guarantee that the available version was the same one registered as “retracted” by the RW database (same venue and publication dates).

Identified “abstracts only” within the conditions above were also recorded, and, in addition, a standard Google Search was conducted, looking for these papers/abstracts in other platforms.

C. Information collection

All searches were performed by two researchers and independently double-checked by a third one. In cases of doubt or disagreement, researchers convened to meet consensus. The following variables were collected for the study: date of article publication, date of retraction, title, and (Google Scholar) citations received. Citations were divided as *post* and *pre-retraction*, where post-retraction citations were defined as those received from the calendar year following retraction onward. From these data, the percentage of post-retraction citations was calculated as: post-retraction citations / (pre-retraction + post-retraction citations).

III. RESULTS

A total of sixty-six (66) retractions could be selected from the Retraction Watch database according to the described criteria. Out of these, 22 (1/3) were available unmarked at the platforms (13 Academia, 5 Research Gate, 4 in both). Additionally to these 22 papers, 07 other un-

marked abstracts were found (04 Research Gate, 03 in other repositories).

Figure 1 shows the pre-post retraction rates of the analyzed papers. Total post-year of retraction citations for the full papers were 730 (average 33.2), and for the abstracts were 166 (average 23.7). Table 1 provides a quantitative summary of the availability and citation data for the 22 full-text articles found unmarked on the platforms, complementing the individual-level visualization presented in Figure 1. Across these articles, citations accumulated both before and after the year of retraction, with the post-retraction period accounting for a substantial proportion of total citations in several cases. The seven abstracts identified without retraction labeling are reported separately, given that their availability was partial and their citation counts represent a distinct subset of the overall findings

Table 1. Summary of availability and citation data for retracted articles found unmarked on academic sharing platforms (2010–2020).

Characteristic	Full papers (n = 22)	Abstracts only (n = 7)
Platform distribution		
Academia.edu only	13	—
ResearchGate only	5	4
Both platforms	4	—
Other repositories	—	3
Total citations (pre + post retraction year)		
Total	1.332	279
Median per article	60,5	39,9
Median per article (IQR)	46,5 (24-75)	23,0 (12-76)
Range	0-320	0-130
Pre-retraction year citations		
Total	602	113
Median per article	27.3	16.1
Median per article (IQR)	16.0 (6–31)	2.0 (0–36)
Range	0–134	0–39
Post-retraction year citations		
Total	730	166
Mean per article	33.2	23.7
Median per article (IQR)	17.0 (6–36)	13.0 (11–35)
Range	0–186	0–61
Proportion of citations post-retraction (%)		
Mean (%)	53.4	74.4
Median % (IQR)	51.0 (36.1–66.0)	78.8 (59.3–93.3)
Range	11.7–100.0	36.8–100.0

IQR: interquartile range. Citation counts retrieved from Google Scholar.

A Spearman correlation between time from publication to retraction (in days) and number of post-retraction citations indicated no statistically significant association ($\rho = -0.108$, $p = 0.631$).

IV. DISCUSSION

A retraction represents the nullification of a scientific paper, implying in its withdrawal from the valid literature. In this study, retractions were restricted to those pertaining to Brazilian first-author affiliations, a pragmatic delimitation adopted to maintain dataset homogeneity and manageability within the scope of the study. The focus on health sciences reflects the direct implications of retracted biomedical literature for clinical practice and public health policies. These choices limit the direct generalizability of findings to other national contexts or scientific disciplines (see “limitations” below).

High citation rates after the retraction of a paper are a well-known phenomenon in the scientific literature [6,7,11-13]. Indeed, Gabehart [14] studied the persistence of citations in retracted articles and concluded that around 30% of their citations occurred after the retraction date, demonstrating that, frequently, researchers continue to reference them without knowledge of their status.

An explanation for this phenomenon is that the indexing of scientific articles is many times deficient, introducing problems to the clear identification of their scientific status. Such flaws have been found even in renowned indexing databases and repositories, such as PubMed and Web of Science [3,4,6,15].

Another possibility, explored here, is that they can still be found over the Internet, without proper signposting, that is, with no indication that a retraction had happened. This was the possibility studied here, and indeed it could be observed that a large proportion of those retracted articles available in the analyzed databases had no indication whatsoever that they were not valid scientific literature anymore.

Worse yet, these retracted papers had a post-retraction citation rate of about 33 citations/paper. For comparison, Pfeifer and Mouton [17] and Wang et al. [18] reported post-retraction citation rates in of 733/82 and 954/128, respectively, values largely exceeded by the rates observed here. Although direct comparisons are limited by differences in study design, sample selection criteria, contexts and time frames, it is therefore possible that the lack of identification contributed for the dissemination and use of retracted material in other scientific reports.

The problem is even more serious if one considers that the analyzed retractions concerned health sciences, and, if left unchecked, may contribute to the incorporation of invalidated findings into literature reviews. This presents potential downstream implications for clinical guidance and evidence-based policy, a scenario that, while plausible, was not directly assessed in this study and warrants dedicated empirical investigation. Besides, time and resources will be diverted to replicate flawed research, adding to the already

existing waste of resources introduced by shoddy scientific literature [16].

One could argue about who could be responsible for the persistent presence of unidentified retracted material on publicly available databases: authors or databases. Indeed, authors have a responsibility to, after a retraction, add clear indications that it has happened to impede the unwarned access to that material. However, databases also have their share of responsibility. Although a formal comparative analysis of their retraction-signaling policies falls outside of the scope of this study, ResearchGate and Academia.edu do not differ much in their governance models and content policies. Both operate as researcher-driven platforms, with some automated content controls and limited editorial vetting; and both would do well striving to develop guidelines and procedures to stop inappropriate dissemination of information.

Limitations: First, as mentioned, the sample is restricted to articles with Brazilian first-author affiliations in health sciences, limiting generalizability to other national or disciplinary contexts. Second, only two platforms were examined; the extent of the problem across other academic repositories, preprint servers, or institutional repositories remains unknown. Third, whether specific retraction reasons (such as data fabrication, fraud, or methodological problems) are associated with greater persistence on sharing platforms or with higher post-retraction citation rates remains an open question. The present dataset, while not powered for such subgroup analyses, may serve as a basis for future investigations examining retraction reason as a potential moderating variable in the persistence of invalidated content. Finally, citation counts were retrieved from Google Scholar, which may differ from other citation databases in coverage and update frequency.

V. CONCLUSION

In summary, this study provides empirical evidence for the high persistence of unmarked retracted publications in academic social networks, highlighting potential noteworthy gaps in retraction-signaling practices on academic social networks. The discussion of this topic addresses basic issues related to academic integrity and the reliability of scientific information. By understanding the factors that contribute to the dissemination of retracted articles and identifying solutions to reduce this problem, it could be possible to promote advances in the way scientific production is managed and disseminated.

Therefore, the availability of these unmarked versions may represent an additional route for the persistence and circulation of retreated content. These findings suggest that current practices for signaling retraction events on academic social networks may be insufficient to prevent continued access to invalidated literature. Potential responses could

include: (a) development of automatic notification mechanisms linking retraction databases (e.g., Retraction Watch, CrossRef) to content-sharing platforms; (b) establishment of clear editorial responsibilities for platform operators regarding content uploaded by registered users; and (c) enhanced guidance for authors regarding the obligation to update or remove uploaded manuscripts following retraction. These observations, while based on a bounded and exploratory sample, may inform future policy discussions on interoperability standards for retraction data.

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CONFLICTS OF INTEREST

The authors declare that they have no conflict of interest relative to the present work.

REFERENCES

- [1]. Barbour V, Kleinert S, Wager E, Yentis S. Guidelines for retracting articles. Committee on Publication Ethics; 2009. <https://doi.org/10.24318/cope.2019.1.4>.
- [2]. Fang FC, Steen RG, Casadevall A. Misconduct accounts for the majority of retracted scientific publications. *Proc Natl Acad Sci U S A* 2012;109:17028–33.
- [3]. Bakker CJ, Reardon EE, Brown SJ, et al. Identification of retracted publications and completeness of retraction notices in public health. *J Clin Epidemiol*. 2024; Sep; 173:111427. doi: 10.1016/j.jclinepi.2024.111427.
- [4]. Salami MO, McCumber C, Schneider J. Analyzing the Consistency of Retraction Indexing. *MetaArXiv Preprints* 2024; Available from: doi.org Access Jan 2025. doi: 10.31222/osf.io/gvfk5.
- [5]. Silva, JAT. Risks associated with the citation of retracted medical literature. *Journal of Health and Social Sciences* 2023; 7(4):363-367. doi: 10.19204/2022/RSKS2.
- [6]. Bolland MJ, Grey A, Avenell A. Citation of retracted publications: A challenging problem. *Accountability in Research* 2022;29:18–25. <https://doi.org/10.1080/08989621.2021.1886933>.
- [7]. Schneider J, Ye D, Hill AM, Whitehorn AS. Continued post-retraction citation of a fraudulent clinical trial report, 11 years after it was retracted for falsifying data. *Scientometrics* 2020; 125:2877–913. <https://doi.org/10.1007/s11192-020-03631-1>.
- [8]. Almeida, Renan Moritz VR et al. Plagiarism allegations account for most retractions in major Latin American/Caribbean databases. *Science and engineering ethics*, v. 22, p. 1447-1456, 2016. <https://link.springer.com/article/10.1007/s11948-015-9714-5>
- [9]. Kintisch E. Is ResearchGate Facebook for science? 2021. <https://doi.org/10.1126/science.caredit.a1400214>.
- [10]. Academia. Academia.edu | About. Academia.Edu 2023. <https://www.academia.edu/about>
- [11]. Piller C. Disgraced COVID-19 studies are still routinely cited. *Science* 2021;371:331
- [12]. Schmidt, M. Why do some retracted articles continue to get cited? *Scientometrics* 129, 7535–7563 (2024).
- [13]. Retraction Watch. Top 10 most highly cited retracted papers. Retraction Watch. <https://retractionwatch.com/the-retraction-watch-leaderboard/top-10-most-highly-citedretracted-papers/> (accessed August 28, 2026).
- [14]. Gabehart, Mary E. An analysis of citations to retracted articles in the scientific literature. 2005. Available <https://ils.unc.edu/MSPapers/3050.pdf>
- [15]. Ortega, José Luis; Delgado-Quiros, Lorena. The indexation of retracted literature in seven principal scholarly databases: a coverage comparison of dimensions, OpenAlex, PubMed, Scilit, Scopus, The Lens and Web of Science. *Scientometrics*, v. 129, n. 7, p. 3769-3785, 2024.
- [16]. Moher, D., Shamseer, L., Cobey, K. et al. Stop this waste of people, animals and money. *Nature* 549, 23–25 2017.
- [17]. Pfeifer MP. The Continued Use of Retracted, Invalid Scientific Literature. *JAMA* 1990;263:1420.
- [18]. Wang Z, Shi Q, Zhou Q, Zhao S, Hou R, Lu S, et al. Retracted systematic reviews continued to be frequently cited: a citation analysis. *Journal of Clinical Epidemiology* 2022;149:137–45.

Figure 1: Pre-/post year of retraction citations for 22 articles available on research-sharing platforms with no retraction identification, 2010-2020

